

RACINE BATTEN, WI, USA

WMO: 726424

Lat: 42.761N

Lon: 87.814W

Elev: 674

StdP: 14.34

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 94818

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.0	2.4	-12.5	2.9	-0.9	-8.3	3.6	3.7	24.0	25.6	21.5	27.3	9.4	280	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	88.7	74.0	85.5	72.5	82.4	70.7	76.4	85.5	74.6	82.5	72.9	80.3	11.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.2	126.6	81.6	72.1	121.6	80.0	70.3	114.5	77.9	40.2	85.8	38.4	82.6	36.9	80.3	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
22.9	20.0	18.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-7.7	93.8	5.9	3.9	-11.9	96.6	-15.4	98.8	-18.7	101.0	-23.0	103.9		
			-8.5	79.0	5.8	2.1	-12.6	80.5	-16.0	81.7	-19.2	82.9	-23.4	84.4		
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.4	23.3	25.1	35.2	45.5	55.8	66.1	72.2	70.9	64.2	52.0	40.1	28.5	
	DBStd	19.23	11.20	10.52	9.99	8.31	8.41	7.44	6.08	5.18	7.41	8.42	9.47	10.56	
	HDD50	3285	828	698	473	186	34	0	0	0	2	78	318	668	
	HDD65	6719	1293	1118	925	588	307	76	10	9	104	410	747	1132	
	CDD50	2685	0	0	13	51	214	483	688	647	426	141	21	1	
	CDD65	644	0	0	0	2	23	109	233	191	78	8	0	0	
	CDH74	4603	0	0	2	45	189	840	1853	1171	460	43	0	0	
	CDH80	1277	0	0	0	7	35	232	599	296	103	5	0	0	
Wind	WSAvg	9.0	9.9	9.9	9.9	10.7	9.2	7.9	7.5	7.3	8.0	9.1	9.5	9.6	
Precipitation	PrecAvg	33.6	1.7	1.3	2.4	3.6	3.3	3.7	3.5	3.6	3.6	2.4	2.7	1.9	
	PrecMax	47.4	4.7	3.7	7.6	7.8	13.5	8.9	8.1	8.4	9.0	7.2	6.7	5.2	
	PrecMin	22.6	0.2	0.0	0.4	0.9	0.5	0.8	1.4	0.4	0.0	0.2	0.6	0.2	
	PrecStd	5.5	1.1	0.9	1.4	1.6	2.4	1.9	1.5	1.9	2.3	1.5	1.5	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.5	55.8	69.9	80.8	83.9	90.4	95.5	91.0	87.3	79.3	67.0	57.1	
		MCWB	48.9	48.2	60.0	62.7	67.9	73.7	77.0	75.6	72.6	67.1	58.3	51.9	
	2%	DB	45.4	48.4	61.3	72.0	79.2	86.3	90.0	86.5	82.7	73.1	62.2	49.9	
		MCWB	40.7	42.7	54.8	59.2	65.6	71.5	75.5	73.5	70.4	63.9	53.8	45.8	
	5%	DB	40.7	42.8	54.9	64.8	74.9	82.0	86.0	83.0	79.3	69.6	57.4	45.4	
		MCWB	37.7	38.1	47.6	54.8	63.7	69.9	73.4	71.6	68.5	60.7	51.5	42.6	
	10%	DB	37.6	38.3	48.9	59.3	70.8	79.1	82.8	80.4	75.4	64.9	54.0	42.4	
		MCWB	35.4	35.2	43.1	50.4	60.9	67.9	71.6	70.0	67.2	58.3	48.5	39.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.9	50.0	60.3	65.8	71.0	77.0	80.0	77.6	75.2	69.9	59.5	53.9
			MCDB	52.3	55.0	67.4	76.6	80.7	85.7	90.3	88.3	83.6	77.2	64.6	56.8
2%		WB	41.7	42.5	55.2	60.2	67.6	73.8	77.2	75.3	72.6	65.9	55.6	46.3	
		MCDB	44.4	47.4	60.9	70.3	75.6	83.1	86.8	82.9	79.7	70.2	60.4	48.1	
5%		WB	37.9	38.4	48.3	56.0	65.3	71.6	75.2	73.6	70.7	62.4	52.2	42.9	
		MCDB	40.3	42.0	53.4	64.6	73.1	80.4	83.5	80.6	76.9	67.2	56.7	45.4	
10%		WB	35.4	35.3	43.6	51.2	62.4	69.2	73.5	71.9	68.6	59.4	48.7	39.3	
		MCDB	37.9	38.1	48.1	57.9	69.5	76.7	81.0	78.0	74.5	64.7	53.5	41.9	
Mean Daily Temperature Range	5% DB	MDBR	12.8	13.5	13.9	15.5	16.7	16.7	16.0	14.8	15.9	15.0	13.4	12.2	
		MCDBR	15.3	17.6	21.8	26.1	23.4	20.4	19.1	18.1	19.3	20.4	18.4	14.8	
		MCWBR	12.9	13.3	16.0	16.3	13.6	10.8	9.2	8.6	9.7	12.5	13.5	13.0	
	5% WB	MCDBR	14.3	16.7	20.8	25.1	21.6	18.9	17.5	15.6	17.0	17.3	16.7	13.7	
		MCWBR	12.7	13.4	16.5	16.9	13.9	11.3	9.6	8.4	9.5	12.9	13.4	12.8	
Clear-Sky Solar Irradiance	taub	0.292	0.311	0.343	0.391	0.416	0.436	0.432	0.424	0.395	0.349	0.312	0.291		
	taud	2.447	2.393	2.394	2.292	2.275	2.270	2.280	2.301	2.357	2.480	2.524	2.490		
	Ebn at Noon	270	282	285	278	273	267	266	265	264	265	262	261		
	Edh at Noon	23	30	33	40	42	42	41	39	35	27	22	21		
All-Sky Solar Radiation	RadAvg	463	747	1126	1454	1680	1915	1948	1681	1361	876	545	380		
	RadStd	62	88	99	124	133	128	88	104	115	100	53	42		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page